MIL-C-24643/56B
3 October 1986
SUPERSEDING
MIL-C-0024643/56A(SH)
8 November 1985
MIL-C-24643/56
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2SWL-7
(INCLUDING VARIATION LS2SWLA-7)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LS2SWL-7 (unarmored) and LS2SWLA-7 (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, tin coated, AWG no. 16, class B stranding.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Nominal thickness 0.019 inch. Identification method 3.
Fourth - Two insulated conductors (one black, one white), cabled together with a nominal lay of 3 inches, to form a pair. At manufacturer's option, a binder tape applied helically with overlap.
Fifth - Braided shield of tin coated copper, size either AWG no. 34 or AWG no. 36. Braid angle 30 to 35 degrees, minimum coverage of 85 percent.
Sixth - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2, on inner tape.

(B) denotes changes.

AMSC N/A FSC 6145
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Seventh - Seven shielded pairs cabled together with a lay not greater than 18 times the overall diameter of the layer. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers shall be employed as necessary to form a firm, well-rounded assembly.
Eighth - Binder tape applied helically with overlap.

Construction variant for type LS2SWL-7

Ninth - Cross-linked polyolefin jacket with cable surface marking. Minimum jacket thickness shall be 0.060 inch. Cable overall diameter shall be 0.860 minimum, 0.910 maximum.

Construction variant for type LS2SWLA-7

(B) Ninth - Cross-linked polyolefin jacket. Minimum jacket thickness shall be 0.060 inch. Cable overall diameter shall be 0.860 minimum, 0.910 maximum.
Tenth - Braided metal armor and paint.

NOTES: 1. Variant LS2SWL-7 military part number - M24643/56-01UN.
2. Variant LS2SWLA-7 military part number - M24643/56-02AN.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	4.66
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield	500
Shield to water	500
Insulation resistance - megohms/1000 feet, minimum	500
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-24643 for limits of water leakage	No failure
Mutual capacitance - at 1 kHz, pF/ft, maximum	35.0
Capacitance unbalance - percent, maximum	10
Group B:	
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250

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EXAMINATION AND TESTS: (Continued)

	Requirements
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum.....	80
Elongation - percent of unaged, minimum..	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum.....	60
Elongation - percent of unaged, minimum..	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum.....	50
Elongation - percent, minimum.....	50
Shrinkage.....	No failure
Permanence of printing (jacket) type LS2SWL-7 - cycles, minimum.....	250
Cable filler removability.....	No failure
Heat distortion - percent of unaged, maximum.....	30
Armor - conformance to materials, construction and coverage (type LS2SWLA-7).....	No failure
Shield - conformance to material, construction and coverage.....	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable) (125 ± 5°C).....	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum.....	75
Acid gas equivalent - percent, maximum	
Jacket.....	2
Fillers.....	2
Insulation.....	18
Halogen content - percent, maximum	
Jacket.....	0.2
Fillers.....	0.2
Flame propagation (cable).....	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum.....	50
Elongation - percent of unaged, minimum.....	50
Smoke index, maximum	
Jacket.....	25
Fillers.....	--
(B) Insulation.....	45

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QUALIFICATION INSPECTION: (Continued)

	Requirements
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTH: 1500 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-56)

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